DSC 210, Fall 2026 — Homework 2 Nicholas Wendt (200478549)

Question 1: Gauss elimination (20 points)

Use Gauss elimination to solve the following equations for $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^\top$:

$$\begin{aligned} x_1 + 5x_2 + 3x_3 &= 1 \\ 2x_1 + 4x_2 - 3x_3 &= -1 \\ -2x_1 + 2x_2 + 3x_3 &= -5 \end{aligned}$$

Solution: First set up an augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 5 & 3 & 1 \\ 2 & 4 & -3 & -1 \\ -2 & 2 & 3 & -5 \end{array}\right]$$

Now solve the system $Ax = b$ by row reducing the matrix.
Apply the row operations $2r_1 - r_2 \to r_2$ and $2r_1 + r_3 \to r_3$

$$\left[\begin{array}{ccc|c} 1 & 5 & 3 & 1 \\ 0 & 6 & 9 & 3 \\ 0 & 12 & 9 & -3 \end{array}\right]$$

Now use row 2's pivot column $2r_2 - r_3 \to r_3$

$$\left[\begin{array}{ccc|c} 1 & 5 & 3 & 1 \\ 0 & 6 & 9 & 3 \\ 0 & 0 & 9 & 9 \end{array}\right]$$

Now use back substitution to solve for $\mathbf{x}$ Starting with the last row we get the equation $9x_3 = 9$ so $x_3 = 1$.
Now the second row gives us the equation $6x_2 + 9x_3 = 3 \to 6x_2 + 9 = 3$. So we see $x_2 = -1$.
Finally, the first equation gives us $1x_1 + 5x_2 + 3x_3 = 1 \to x_1 - 5 + 3 = 1$. So we see $x_1 = 3$.

$$\mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^\top = \begin{bmatrix} 3 & -1 & 1 \end{bmatrix}$$

Question 2: LU decomposition (20 points)

Perform LU decomposition for the matrix corresponding to the equations in **Question 1**. Using the matrices L and U, do forward and backward substitution to solve for $\mathbf{x}$. Match your solution with that of **Question 1**.

Solution:

$$A = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 4 & -3 \\ -2 & 2 & 3 \end{bmatrix}$$

We can factor A into a lower triangular matrix (L) and an upper triagular matrix (U) such that $A = LU$

We row reduce A to an upper triangular matrix without using row interchanges.

We keep track of the scalers in L.

First 2 operations $r_2 - 2r_1$, $r_3 + 2r_1$. We keep track of the scalers of the pivot row in L according

to $r_i - cr_j$ where c is the row i column j entry in L.

Now we have:

$$A = \begin{bmatrix} 1 & 5 & 3 \\ 0 & -6 & -9 \\ 0 & 12 & 9 \end{bmatrix} \quad , \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & _ & 1 \end{bmatrix}$$

The last row operation is $r_3 + 2r_2$. Since A in now upper triangular, we write U

$$U = \begin{bmatrix} 1 & 5 & 3 \\ 0 & -6 & -9 \\ 0 & 0 & -9 \end{bmatrix} \quad , \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

Firs use forward substitution to solve $Ly = b$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & -2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -5 \end{bmatrix}$$

$$\begin{aligned} y_1 &= 1 \\ 2y_1 + y_2 = -1 &\implies y_2 = -1 - 2(1) = -3 \\ -2y_1 - 2y_2 + y_3 = -5 &\implies y_3 = -5 + 2(1) + 2(-3) = -9 \end{aligned}$$

So $\mathbf{y} = \begin{bmatrix} 1 \\ -3 \\ -9 \end{bmatrix}$.

Then use backward substitution to solve $Ux = y$:

$$\begin{bmatrix} 1 & 5 & 3 \\ 0 & -6 & -9 \\ 0 & 0 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ -9 \end{bmatrix}$$

$$
\begin{aligned}
-9x_3 = -9 &\implies x_3 = 1 \\
-6x_2 - 9x_3 = -3 &\implies x_2 = \frac{-3 + 9(1)}{-6} = -1 \\
x_1 + 5x_2 + 3x_3 = 1 &\implies x_1 = 1 - 5(-1) - 3(1) = 3
\end{aligned}
$$

$$
\mathbf{x} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}
$$

This matches the solution from question 1.

Question 3: QR decomposition (20 points)

Perform QR decomposition for the matrix corresponding to the equations in **Question 1** using the Gram-Schmidt algorithm. Using the decomposition, solve for **x**. Match your solution with that of **Question 1** and **Question 2**.

Solution:

$$A = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 4 & -3 \\ -2 & 2 & 3 \end{bmatrix}$$

First find q_1. $\quad q_1 = \frac{a_1}{\|a_1\|}$

$\|a_1\| = \sqrt{1+4+4} = 3$. So, $q_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ -2/3 \end{bmatrix}$.

We record $\|a_1\|$ as the r_{11} entry in R.

Next we calculate $a_2^\perp = a_2 - \langle a_2, q_1 \rangle q_1$. $\quad \langle a_2, q_1 \rangle = \frac{5}{3} + \frac{8}{3} - \frac{4}{3} = \frac{9}{3} = 3$. This is the r_{12} entry in R.

Then $a_2^\perp = a_2 - 3q_1 = \begin{bmatrix} 5 \\ 4 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 4 \end{bmatrix}$. $\quad$ Then $\|a_2^\perp\| = \sqrt{16+4+16} = 6$. $\|a_2^\perp\|$ is the r_{22} entry in R and $q_2 = \frac{a_2^\perp}{\|a_2^\perp\|}$. So

$$q_2 = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$$

Now onto the last vector for the orthonormal basis. $a_3^\perp = a_3 - \langle a_3, q_1 \rangle q_1 - \langle a_3, q_2 \rangle q_2$.

$\langle a_3, q_1 \rangle = 1 - 2 - 2 = -3$. This is the r_{13} entry in R. $\langle a_3, q_2 \rangle = 2 - 1 + 2 = 3$. This is the r_{23} entry in R.

$$a_3^\perp = \begin{bmatrix} 3 \\ -3 \\ 3 \end{bmatrix} - \begin{bmatrix} -1 \\ -2 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix}$$

Then $\|a_3^\perp\| = \sqrt{4+4+1} = 3$. This is the r_{33} entry in R.

$$q_3 = \frac{a_3^\perp}{\|a_3^\perp\|} = \begin{bmatrix} 2/3 \\ -2/3 \\ -1/3 \end{bmatrix}$$

We have now factored A into Q and R.

$$A = QR = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ -2/3 & 2/3 & -1/3 \end{bmatrix} \begin{bmatrix} 3 & 3 & -3 \\ 0 & 6 & 3 \\ 0 & 0 & 3 \end{bmatrix}$$

Question 4: Gram-Schmidt process (20 points)

Show that the residual vector $\mathbf{a}_i^{\perp}$ is orthogonal to $\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_{i-1}$ in the Gram-Schmidt process.

Solution: Given a set of linearly independent vectors $a_1, a_2, ..., a_n$ then for each i

$$a_i^{\perp} = a_i - \sum_{k=1}^{i-1} (q_k^{\top} a_i) q_k \quad , \quad q_i = \frac{a_i^{\perp}}{\|a_i^{\perp}\|}$$

We claim for each i, $q_j a_i^{\perp} = 0$ for all $j = 1, 2, ..., i-1$. i.e. all the vectors beofre column i are orthogonal to column i.

Base case: For $i = 1$ there are no earlier $q_j's$ we need to make column 1 orthogonal to, so $a_1^{\perp} = a_1$ and $q_1 = \frac{a_1^{\perp}}{\|a_1^{\perp}\|}$.

Induction step: Assume $q_1, q_2, ..., q_{i-1}$ are orthonormal so that

$$q_j^{\top} q_k = \delta_{jk} = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases} \qquad \text{for } j, k \leq i-1$$

Fix any $j \in 1, 2, ..., i-1$. Then

$$\begin{aligned} q_j^{\top} a_i^{\perp} &= q_j^{\top} \left(a_i - \sum_{k=1}^{i-1} (q_k^{\top} a_i)\, q_k \right) \\ &= q_j^{\top} a_i - \sum_{k=1}^{i-1} (q_k^{\top} a_i)\, (q_j^{\top} q_k) \quad \text{(linearity; } q_k^{\top} a_i \text{ is a scalar)} \\ &= q_j^{\top} a_i - \sum_{k=1}^{i-1} (q_k^{\top} a_i)\, \delta_{jk} \qquad \text{(inductive hypothesis)} \\ &= q_j^{\top} a_i - q_j^{\top} a_i \\ &= 0. \end{aligned}$$

So $a_i^{\top}$ is orthogonal to every q_j where $j < i$. Since the preecding columns are assumed to be linearly independent of column i, $a_i^{\perp} \neq 0$ since it can't be in the span$(q_1, q_2, ..., q_{i-1})$.

Orthonormality is preserved since $a_i^{\perp} \neq 0$, then $q_i = a_i^{\perp} / \|a_i^{\perp}\|$ has unit length. Dividing by a positive scalar doesn't affect orthogonality, so $q_j^{\top} q_i = 0$ for all $j < i$. ■

Question 5: Rank deficient matrices (20 points)

Compute the QR decomposition of the following two matrices using the Gram-Schmidt process and properties of the matrix $\mathbf{Q}$.

(i) $\mathbf{A} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & -1 & 1 \end{bmatrix}$

(ii) $\mathbf{B} = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ -1 & 1 & -1 \end{bmatrix}$

Note: the matrices $\mathbf{Q}$ and $\mathbf{R}$ must be square matrices.

Solution:

(i)

(ii)